Artificial Intelligence for Climate and Weather Forecasting: A Systematic Literature Review Using the PRISMA Methodology (2019–2024)

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Abstract

However, for almost half a century now, the field of weather and climate prediction has been dominated by NWP models which make extensive use of physical simulations of atmospheric dynamics and run on powerful supercomputers. Within the past five years, however, there has been a revolution as deep learning techniques were shown to be able to match or even exceed the performance of NWP models at a fraction of their computational costs. The purpose of this paper is to provide a Systematic Literature Review of 101 publications related to the topic and released between 2019 and 2024 according to the PRISMA 2020 protocol. The search of academic literature was performed in six relevant online databases, and based on carefully crafted inclusion/exclusion criteria, we identified a body of literature worthy of review. This paper contains a discussion of five key research questions about the field. Our observations make it quite evident that Transformer models and GNNs now reign supreme in their respective domains, with ERA5 reanalysis serving as the de facto training dataset for both. In addition, foundation models such as GraphCast, Pangu-Weather, FourCastNet, FengWu, NeuralGCM, and ClimaX have surpassed a certain benchmark in terms of their accuracy compared to NWP models. However, there is still much work to be done in the field. Forecasting extremely rare phenomena, making the models interpretable, ensuring proper calibration of probabilities, and covering developing countries with less data availability are all aspects that need attention.

Keywords: artificial intelligence, deep learning, weather forecasting, climate modelling, transformer, graph neural network, LSTM, foundation model, PRISMA, systematic literature review, numerical weather prediction, extreme weather, ensemble forecasting, probabilistic forecasting

1. Introduction

Accurate weather and climate forecasting is one of the most practically important scientific problems of our time. Billions of people rely on weather forecasts every day for decisions ranging from simple daily plans to emergency evacuations, agricultural scheduling, and infrastructure management. At the same time, the climate is changing in ways that make extreme weather events more frequent, more severe, and harder to predict using tools developed for more stable historical conditions [1].

For more than seven decades, the dominant approach to weather prediction has been Numerical Weather Prediction (NWP): solving systems of partial differential equations that represent atmospheric physics on global computational grids. NWP has improved enormously over the years, but it remains very expensive to run, takes considerable time to produce a forecast, and has difficulty capturing rare extreme events accurately [2, 3].

Artificial Intelligence, and deep learning in particular, began to seriously challenge NWP around 2022. Within a few years, multiple independently developed AI models demonstrated that a data-driven system trained on decades of reanalysis data could match or outperform the world's best NWP systems on standard benchmark metrics, while producing forecasts in seconds rather than hours on much cheaper hardware [4, 5, 6, 7].It is much more than a marginal advance in one particular test case — it marks a real change in the way the field approaches prediction.

The rapid development in the field of AI weather research requires some careful analysis. For an early researcher interested in exploring this area, there will be hundreds of papers available in academic journals as well as at conferences related to meteorology, computer science, climate sciences, and engineering. The lack of a clear framework for organizing this wealth of literature makes it hard to distinguish between the established knowledge, the disputed ideas, and the open problems in this field.

This paper seeks to provide such a framework using Systematic Literature Review (SLR) based on PRISMA 2020 method [8]. Our search of six leading scientific databases produced 101 selected articles to answer five research questions.

1.1 Research Questions

Five research questions (RQs) guide this review:

RQ1: What types of AI and deep learning models have been used for climate and weather forecasting from 2019 to 2024, and what changes or improvements have been made to these models over time?

RQ2: To what extent are the performances of AI-based weather prediction models comparable with those of conventional NWP models concerning accuracy, efficiency, and scalability?

RQ3: Which publicly available datasets and evaluation metrics are most commonly used by researchers when testing AI models for weather and climate forecasting?

RQ4: What weather and atmospheric variables (such as temperature, humidity, wind speed, pressure, etc.) are most often used as input features in AI-based forecasting models?

RQ5: What are the main challenges, limitations, and possible future research directions for using AI in climate and weather forecasting?

2. Background and Related Literature Reviews

2.1 A Brief History of AI in Weather Forecasting

ANNs were first deployed in the domain of weather forecasting in the early 1990s, with applications including local temperature and precipitation estimation. These models were shallow, handcrafted feature-based, and unable to compete with numerical weather predictions in scale. This situation changed dramatically with the advent of modern deep learning.

LSTM networks began to be employed in weather time-series forecasting after 2015 [9]. Convolutional LSTMs (ConvLSTM) [10] represented a straightforward solution to modeling spatiotemporal sequences such as radar images, thus enabling the possibility of precipitation nowcasting using data-driven methods. TrajGRU [11] is an improvement over ConvLSTM because it learns to capture motion patterns instead of using static convolution kernels.

In the years 2019-2021, there have been organized proofs of competitive accuracy of deep learning models to NWP on regional and global scales. In particular, Ham et al. [12] proved the possibility of making a forecast of El Niño conditions 18 months before its appearance using a CNN trained solely on climate model output with superior results compared to traditional dynamical seasonal prediction systems. The work [13] proposed DeepSD, an application of the super-resolution CNNs approach to climate downscaling. Finally, Weyn et al. [14] demonstrated how one can utilize deep convolutional networks running on a cubed sphere grid for improving global data-driven forecasting.

The years 2022–2024 brought the foundation model era. Pangu-Weather [4], GraphCast [5], FourCastNet [6], FengWu [7], and NeuralGCM [15] all demonstrated forecasting performance that rivals ECMWF's operational Integrated Forecast System (IFS), representing a qualitative change in the state of the art.

2.2 Existing Reviews and Identified Gaps

Table 1 summarises the most relevant existing reviews and identifies the specific gaps that motivated this work.

Table: Comparison of existing systematic and narrative reviews on AI for weather and climate forecasting.

Review	Year	Main Focus	Gap / Limitation
Bochenek & Ustrnul (2022)	2022	ML methods across weather variables; text-mining of 500 papers	Not PRISMA-based; lacks architectural taxonomy; no foundation model coverage
Dueben & Bauer (2022)	2022	Challenges in ML-based global weather and climate models	Focuses on design challenges; does not systematically compare AI vs. NWP performance
Rasp et al. (2024)	2024	WeatherBench 2 benchmark for data-driven global models	Benchmark paper, not an SLR; limited coverage of extreme events and local methods
Zemnazi et al. (2024)	2024	ML and DL approaches for weather forecasting: a review	Narrative review without PRISMA; limited coverage of post-2022 foundation models
This Review (2025)	2025	AI for climate & weather: PRISMA-based, 101 papers, 2019–2024	Addresses all five gaps identified in the paper

From the above discussion, it can be observed that five key gaps were identified and served as the motivation behind the current review, including

- Gap 1: Lack of benchmarking. It was observed that most review papers did not compare existing AI models based on the standard datasets and evaluation metrics [2].

- Gap 2: Limited focus on rare events. Most reviews focused on the average weather conditions and failed to emphasize the importance of the forecasting performance in rare events like floods, heatwaves, rapid cyclone intensification, and more [16].

- Gap 3: Coverage gap. There is very little information regarding the post-2022 explosion in the development and applications of large-scale foundation models in weather forecasting.

- Gap 4: Explainability. Most review papers do not evaluate the physical explanation and interpretability of AI weather forecasts.

- Gap 5: Insufficient focus on developing countries. The majority of review papers used datasets from Europe and North America, and AI forecasts in developing regions, such as South Asia and Africa, were poorly covered.

3. Methodology

This review follows the PRISMA 2020 (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines [8]. The complete methodology is described below.

3.1 Search Query

The following Boolean search string was applied consistently across all six databases:

("artificial intelligence" OR "machine learning" OR "deep learning" OR "neural network" OR "transformer" OR "LSTM" OR "CNN" OR "GRU" OR "foundation model") AND ("climate forecasting" OR "weather prediction" OR "weather forecasting" OR "numerical weather prediction" OR "climate modeling" OR "precipitation forecasting" OR "temperature prediction" OR "extreme weather") AND ("accuracy" OR "performance" OR "generalization" OR "real-time" OR "downscaling" OR "ensemble" OR "uncertainty quantification") AND ("2019" OR "2020" OR "2021" OR "2022" OR "2023" OR "2024")

3.2 Databases Searched

Papers were retrieved from six sources: (1) Google Scholar, (2) ACM Digital Library, (3) Springer Nature Link, (4) Elsevier ScienceDirect, (5) IOS Press, and (6) Taylor & Francis Online. Only freely downloadable papers — open-access, arXiv pre-prints, or author-shared manuscripts — were retained.

3.3 Inclusion and Exclusion Criteria

Table 2 presents the formal inclusion and exclusion criteria applied at each PRISMA stage.

Table: Inclusion and exclusion criteria applied during the PRISMA screening process.

ID	Inclusion Criteria	ID	Exclusion Criteria
IC1	Published 2019–2024 (landmark 2017–2018 papers with ≥200 citations retained as baselines)	EC1	Not related to AI/ML for weather or climate
IC2	Uses at least one AI or DL architecture (LSTM, CNN, GRU, Transformer, GNN, etc.)	EC2	Published before 2019, except landmark architecture papers used as reference
IC3	Applied to climate/weather forecasting, nowcasting, downscaling, or extreme weather	EC3	Paywalled or not freely downloadable at time of collection
IC4	Reports quantitative performance metrics (RMSE, MAE, ACC, CRPS, or equivalent)	EC4	Duplicate publication — highest-quality venue retained
IC5	Peer-reviewed journal or conference paper, or credible pre-print from identifiable group	EC5	Foundational DL papers (ResNet, ViT, etc.) with no direct weather application
IC6	Freely downloadable: open-access, arXiv, or author-posted PDF	EC6	Domain is purely energy, finance, or NLP with no atmospheric component

3.4 PRISMA Flow

Figure 1 (represented below as a structured table) shows the complete PRISMA selection process. Starting from approximately 180 database records, duplicates were removed and title/abstract screening excluded papers clearly outside scope. Further elimination was done by full-text review of 108 articles, after which 7 more were excluded.

IDENTIFICATION Records retrieved from 6 databases: n ≈ 180	Duplicates removed: n ≈ 35

↓

SCREENING Title & abstract screening: n = 145	Excluded (non-AI / paywalled / off-topic): n = 37

↓

ELIGIBILITY Full-text reviewed: n = 108	Excluded at full-text (no metrics / wrong domain / paid): n = 7

↓

INCLUDED Final active reading list: n = 101 papers	✓ Included

Figure 1. PRISMA 2020 flow diagram showing paper selection across six databases. The final included set consists of 101 papers.

4. Taxonomy and Classification of AI Architectures (RQ1)

4.1 Overview of Architecture Families

Analysis of the 101 included papers reveals six distinct families of AI architecture applied to weather and climate forecasting. Table 3 provides a structured taxonomy with representative models and key characteristics.

Table: Taxonomy of AI architecture families applied to weather and climate forecasting in the reviewed literature.

Architecture Family	Dominant Era	Papers (n)	Representative Models	Key Strengths
Transformer-based	2022–2024	14	Pangu-Weather, FengWu, FourCastNet, Earthformer	Long-range dependencies, scalable pre-training
LSTM / GRU / RNN	2019–2022	10	ConvLSTM, TrajGRU, Bi-LSTM, Attention-LSTM	Temporal sequence modelling, weather time-series
CNN / ResNet	2019–2022	8	DeepSD, ENSO-CNN, Cubed-sphere CNN, ClimateNet	Spatial pattern extraction, super-resolution
Graph Neural Networks	2022–2024	6	GraphCast, Keisler GNN	Irregular grids, global teleconnection modelling
Hybrid / Physics-AI	2021–2024	5	NeuralGCM, NowcastNet, O'Gorman & Dwyer	Physical consistency, conservation laws

Architecture Family	Dominant Era	Papers (n)	Representative Models	Key Strengths
Foundation / Multi-task	2022–2024	4	ClimaX, Aurora, MetNet-3	Generalisation, multi-task, multi-variable learning

4.2 Recurrent Architectures: LSTM and GRU

In the years between 2019 and 2021, LSTM networks and its related models have been prevalent in studies on AI for weather. The capability of LSTMs to retain past information across extended periods makes it a natural choice for predicting future trends, considering that present weather relies upon events that occurred many hours ago. For instance, Hewage et al. [9] demonstrated the superior performance of LSTMs and TCNs against the classic model ARIMA. Zenkner and Navarro-Martinez [17] confirmed that Bi-LSTM models achieve strong prediction accuracy for multi-variable meteorological time-series.

The ConvLSTM architecture [10] extended LSTM networks by replacing fully connected operations with convolutions, enabling simultaneous modelling of spatial and temporal dependencies. This made ConvLSTM the dominant approach for precipitation nowcasting for several years. TrajGRU [11] later improved upon ConvLSTM by learning spatially varying motion patterns, leading to better performance on rainfall datasets.

4.3 Convolutional Neural Networks

CNNs are effective at detecting spatially localised patterns in gridded weather data, radar maps, and satellite imagery. Ham et al. [12] used a CNN trained on CMIP5 simulations to predict El Niño 18 months in advance, far exceeding the lead times achievable with dynamical models. DeepSD [13] applied single-image super-resolution methods to climate downscaling, producing high-resolution regional projections from coarse global model output. ClimateNet [18] introduced an expert-labelled dataset for detecting extreme weather events such as tropical cyclones and atmospheric rivers using deep learning segmentation.

4.4 Transformer-Based Models

Transformers became the dominant architecture from 2022 onwards, driven by their ability to model long-range spatial and temporal dependencies through self-attention. The initial transformer [19], for example, incorporated the multi-head self-attention mechanism along with parallelization when dealing with sequence data, which proved to be beneficial for efficient training with massive atmospheric datasets.

On the other hand, Pangu-weather [4] developed the Earth-specific 3D Transformer (3DEST), where the atmospheric pressure level was modeled as another spatial dimension along with latitude and longitude. The model can now learn the structure of the atmosphere horizontally as well as vertically. The Earthformer [20] model suggested the Cuboid Attention Block, where the spacetime cube could be broken into chunks that could be processed in parallel. FengWu [7] uses a multimodal, multitask Transformer with uncertainty-weighted loss functions and a replay-buffer mechanism to extend skillful forecasting beyond 10 days lead time.

4.5 Graph Neural Networks

GNNs represent the atmosphere as a graph where nodes are spatial locations and edges encode atmospheric interactions. GraphCast [5] showed that GNNs on icosahedral grids naturally avoid coordinate-singularity problems near the poles, leading to more uniform spatial accuracy globally. The encode-process-decode structure allows efficient multi-scale message passing across the atmospheric grid. Weyn et al. [14] achieved similar benefits by projecting atmospheric data onto a cubed-sphere geometry.

4.6 Hybrid Physics-AI Models

NeuralGCM [15] integrates learned dynamics modules into a General Circulation Model (GCM) structure, coupling a differentiable dynamical core with neural network physics modules and training the full system end-to-end. The result performs competitively on both deterministic weather forecasting and multi-year climate simulation. O'Gorman and Dwyer [21] showed that machine learning can emulate moist convection parameterisation inside a GCM, though performance degrades outside the training climate distribution. Das et al. [22] combined a physics-embedded deep generative model with NWP baselines for extreme precipitation nowcasting, outperforming both the physics-only and data-only approaches.

4.7 Foundation and Multi-task Models

Foundation models trained on large-scale datasets apply self-supervised learning to generalise across multiple Earth system tasks. ClimaX [23] is trained on ERA5 and CMIP6 climate projections using a self-supervised pre-training approach, and can be fine-tuned for climate projection, air quality prediction, and ocean forecasting tasks. Aurora [24] demonstrates that a single foundation model can handle diverse environmental forecasting tasks including air pollution and ocean wave prediction. MetNet-3 [25] extends this idea to probabilistic day-ahead prediction using both sparse and dense observational inputs.

4.8 Architecture Evolution

The field went through three distinct phases from 2019 to 2024. The period from 2019 until 2021 was characterized by LSTM and CNN model dominance. From 2022 to date, transformer and GNN approaches have become the predominant choice in global forecasting tasks. Large foundation models that utilize training on diverse multi-variable data became prominent after 2023. Each phase improved upon the last phase with the use of residual learning [26] and attention modules being the norm in all recent models.

5. Comparative Analysis: AI vs. NWP (RQ2)

The question remains whether the AI models actually beat the NWP models, which is physics based, and if they do, under what circumstances. From the 101 studies analyzed, the answer that can be drawn is as follows: the AI models are definitively better than physics-based models in deterministic forecast prediction of normal meteorological variables at the medium range.

5.1 Performance on Standard Benchmarks

Table 4 shows the comparison between important AI weather models and ECMWF IFS based on the findings in the articles under discussion and the benchmarks provided by the WeatherBench 2 benchmark [27].

Table: Comparison of major AI weather models versus ECMWF IFS. ✓ = beats IFS; ~ = comparable.

Model	Architecture	Year	Key Metric	Beats IFS?	Inference Speed
IFS HRES (baseline)	Physics NWP	–	Z500 RMSE baseline	–	~60 minutes
FourCastNet	AFNO Transformer	2022	Competitive on select variables	~	<2 seconds
Pangu-Weather	3D Transformer	2022	Outperforms IFS on most lead times	✓	<1 minute
GraphCast	GNN	2023	Outperforms ECMWF HRES on most targets	✓	<1 minute
FengWu	Transformer	2023	Skillful beyond 10 days; beats GraphCast	✓	<1 minute

Model	Architecture	Year	Key Metric	Beats IFS?	Inference Speed
NeuralGCM	Hybrid GCM+NN	2024	Competitive deterministic and ensemble	✓	Much faster than NWP
MetNet-3	Neural network	2023	Day-ahead probabilistic skill	✓	Real-time capable
ClimaX	Foundation model	2023	Strong multi-task generalisation	~	<1 minute

5.2 Where AI Models Excel

Computational speed. AI models run orders of magnitude faster than NWP. GraphCast [5] produces a global 10-day forecast in under one minute on a single GPU. ECMWF's operational IFS requires around one hour on a supercomputer cluster. FengWu [7] also delivers global forecasts in seconds. This speed advantage makes it economically feasible to run hundreds of ensemble members at much lower cost.

Medium-range deterministic accuracy. Multiple independently developed models now outperform IFS for 3–10 day global forecasts of geopotential, temperature, and wind [4, 5, 7]. WeatherBench 2 [27] confirms that several data-driven systems exceed IFS HRES on key deterministic metrics including Z500 RMSE. FengWu in particular extends skillful global forecasting beyond 10 days lead time.

Precipitation nowcasting. ConvLSTM and TrajGRU [10, 11] outperformed optical-flow radar extrapolation methods on standard nowcasting benchmarks. Das et al. [22] showed that a hybrid physics-AI nowcasting model captures convective hotspots better than the HRRR operational NWP model for extreme precipitation events.

Local correction and post-processing. Schulz and Lerch [28] compared eight ML and statistical methods for wind gust post-processing and found that neural network methods substantially outperform traditional EMOS approaches. Neural network calibration of ECMWF temperature and precipitation forecasts using station observations also produced significant improvements [29].

5.3 Where NWP Retains Advantages

Rare and unprecedented extreme events. Pasche et al. [30] specifically evaluated deep-learning forecast models on high-impact extreme events and found that performance degrades for events outside the historical training distribution. Vorndran et al. [16] confirmed this challenge in radiation fog prediction, where rare weather types are difficult to forecast due to data imbalance.

Physical consistency. NeuralGCM [15] reports climate drift under modified boundary conditions. O'Gorman and Dwyer [21] showed that convection parameterisations learned from one climate regime do not extrapolate to warmer or cooler climates. Purely data-driven approaches may predict physically impossible values, for example, humidity below zero or unrealistic pressure distributions.

Forecast realism. The issue of blurring in AI predictions has been highlighted in the WeatherBench 2 work [27] where the authors mention that the AI system could predict accurately with very low RMSE but predicts the smoothed fields without any real physical structures.

Fairness of evaluation. It is important to note here that the authors of FengWu [7] specifically mention that the AI system that utilizes ERA5 analysis cannot be compared to operational NWP systems because the latter must have data assimilation from real observations.

6. Datasets and Evaluation Metrics (RQ3)

6.1 Benchmark Datasets

Table 5 illustrates the datasets that are frequently applied in the studies analyzed in this research paper. The primary reason for the usage of the data in the ERA5 reanalysis is that it covers the entire world and has a historical timeline.

Table: Most commonly used datasets in 100 selected studies (RQ3).

Dataset	Type	Papers (n)	Description and Key Properties
ERA5 (ECMWF)	Reanalysis	34	Global, 0.25° resolution, 1940–present, 80+ variables, freely available
WeatherBench / WB2	Benchmark suite	12	Standardised train/val/test splits from ERA5; enables fair cross-model comparison
CMIP5 / CMIP6	Climate model ensemble	9	Multi-model climate projections; used for pre-training ClimaX and foundation models
NEXRAD / radar datasets	Radar reflectivity	7	High-resolution precipitation; primary source for nowcasting studies
ECMWF HRES / ENS	NWP output	6	Used as reference baseline for post-processing and comparative evaluation

Dataset	Type	Papers (n)	Description and Key Properties
Surface station obs.	In-situ	5	Regional station data for temperature, wind, humidity calibration studies
ClimateNet	Labelled climate events	3	Expert-labelled dataset for atmospheric river and tropical cyclone detection
CloudSat / satellite	Remote sensing	3	Used for snowfall mapping and upper-tropospheric atmospheric characterisation

6.2 Evaluation Metrics

Six evaluation metrics appear consistently across the reviewed papers:

1. RMSE (Root Mean Square Error) is undoubtedly the most prevalent among these, being referenced in over 85% of the analyzed publications. The key feature of this criterion is that it penalizes larger deviations from the target values significantly better than smaller ones, thus being able to detect those critical forecast errors that are vital for practical applications. It should also be noted that it is the main criterion considered in WeatherBench benchmarks [10]. In turn, the Anomaly Correlation Coefficient (ACC) evaluates how accurately the predicted anomaly matches the observations compared to the climatological values. A value higher than 0.6 is traditionally regarded as an indicator of adequate forecast quality in terms of the Z500 parameter. The MAE (Mean Absolute Error) typically comes as an additional measure along with RMSE. [27].

2. ACC (Anomaly Correlation Coefficient): Indicates the correlation between the predicted anomaly and the climatological mean. Scores higher than 0.6 are generally accepted to be indicative of a useful forecast, at least when it comes to predicting Z500.

3. MAE (Mean Absolute Error): Used together with RMSE; less sensitive to extreme values. Recommended for some precipitation prediction studies and nowcasting tasks.

4. In the context of probabilistic predictions, the score of choice is CRPS (Continuous Ranked Probability Score). Smaller scores are associated with better calibrated forecasts; CRPS receives much attention in the post-processing literature [25]. Brier Score is employed when trying to answer another question: How accurate is a given model in estimating the probability of an occurrence of a binary event (such as exceeding some precipitation threshold)? CSI (Critical Success Index) is a measure commonly employed when assessing nowcasting performance; its purpose is similar to that of Brier Score. [28].

5. Brier Score: Measure of the probability of a binary event relevant to questions such as

’will the precipitation exceed the threshold value?’

6. CSI (Critical Success Index): Applied in nowcast evaluation in cases of correctly predicted events of precipitation exceeding a threshold.

There is one commonality among all the results discussed so far, and this point needs to be elaborated. Different metrics are used with different training, evaluation, and geographic scopes between papers. Such inconsistency makes comparison on a quantitative level between studies almost meaningless. However, the initiative taken by WeatherBench 2 [10] to tackle this issue through standardised evaluation data and server should not be overlooked. Even more illustrative is the case with CRPS: in spite of rising interest in uncertainty-based predictions in the field, probabilistic metrics are present in less than 25

7. Key Meteorological Features Used as Predictors (RQ4)

Table 6 summarises the atmospheric variables most commonly used as inputs to AI forecasting models across the 101 reviewed papers.

Table: Frequency of atmospheric input features used across reviewed AI forecasting papers (RQ4).

Feature	Papers (n)	Typical Levels / Source
Geopotential height (Z)	31	500, 850, 1000 hPa
Temperature (T)	30	850 hPa, 2 m surface
Wind speed (u, v components)	27	10 m, 850, 500 hPa
Specific / relative humidity (q)	22	850 hPa, surface
Precipitation / radar reflectivity	19	Surface / radar
Sea surface temperature (SST)	12	Ocean surface
Sea-level pressure (SLP)	10	Surface
Soil moisture	5	Surface / root zone
Lightning parameters	4	Surface station
Aerosol optical depth (AOD)	3	Column integral

500-hPa geopotential height and 850-hPa temperature are two widely used predictors that reflect the tradition of NWP modeling, where the former is the leading factor for steering flow in the middle troposphere while the latter characterizes synoptic air masses. Wind vectors are vital for impact assessments and nowcasting tasks. The humidity predictor is essential for precipitation forecasting, while sea surface temperature (SST) is a key factor in seasonal and ENSO-related forecasts [12].

An interesting development in the context of foundation models is a shift towards employing all variables together as multichannel inputs. For example, Pangu-Weather [4] uses 69 variables, whereas GraphCast [5] incorporates more than 200 variables, compared to early approaches that train models on only 3–6 well-chosen predictors. Modern foundation models like ClimaX [23] and Aurora [24] can process all atmospheric variables together, allowing for joint learning of multiple tasks.

Mostajabi et al. [31] demonstrated that even a reduced set of just four surface variables — temperature, humidity, pressure, and wind — can support useful lightning nowcasting without radar or satellite data. This finding is important for applications in data-sparse regions where advanced observation systems may not be available.

8. Discussion

8.1 What the Literature Establishes

There are three observations that are validated through several independent studies and may be considered established results from the literature review.

First, AI has reached the quality bar for medium-term forecasting. Several independently trained models (Pangu-Weather, GraphCast, FengWu, FourCastNet, NeuralGCM) perform better than ECMWF IFS on traditional metrics when used to forecast the weather for 3 to 10 days into the future [4, 5, 7, 15]. These results were obtained using a variety of benchmark data sets, validation periods, and geographic locations and have been independently confirmed using WeatherBench 2 [27].

Second, AI is an order of magnitude more efficient in terms of computing requirements. While it takes a supercomputing cluster about an hour to generate a 10-day global weather forecast with IFS, the new AI-based model does so in less than a minute on a consumer-grade GPU [5, 15]. This implies that an ensemble forecast can be run

Third, the standardization of ERA5 and transformers has occurred. The use of ERA5 reanalysis data set has been the main choice in most of the reviewed articles as the training data set. In addition to that, from 2022 onwards, the use of transformer models surpassed the popularity of CNN and LSTM models. Graph structures were considered to represent spatial locations on the global grid.

8.2 Active Debates in the Literature

Finally, there are several open issues in the domain of study, which are still being debated within the field.

Replacement vs. augmentation is a major point of contention in the current state of research in deep-learning applications to meteorology. There are works that see progress in AI as proof of concept of its full substitution of traditional NWP for forecasting tasks, whereas others prefer a mixed approach, which is especially evident in the development of NeuralGCM [15]. In post-processing, there is also support for this idea [28, 29].

Another open issue relates to deterministic skills vs. physical realism. It was mentioned by WeatherBench 2 [27] under the term "blurring problem": an artificial intelligence-based prediction model might show low RMSE because it predicts smooth and mean-like fields that cannot mimic the realistic fine structures of the weather field. The issue partially contradicts findings by Das et al. [22], where authors managed to capture the convective structure better than HRRR with their hybrid approach.

The third area of controversy involves fairness in assessment. This is pointed out explicitly by FengWu [7], who observes that the AI models trained using ERA5 reanalysis data cannot be said to be entirely comparable to operational NWP models because the latter need to assimilate observations in real time.

8.3 Critical Unresolved Issues

In addition to established results and ongoing controversies, five open questions are highlighted in this paper.

Forecasting extreme events. Pasche et al. [30] found that deep learning models exhibit worse performance at forecasting extreme events compared to other forecast models. This is an inherent property of AI models trained on historical data statistics – they consistently underestimate rare events beyond the training distribution. Radiation fog prediction models, for instance, suffer from similar issues, according to Vorndran et al. [16].

Consistency with physics. NeuralGCM [15] finds climate drifts when running simulations with adjusted boundary conditions. O'Gorman and Dwyer [21] found that convective parameterizations were incapable of predicting behaviour for climates outside their temperature ranges.

Interpretability and Trustworthiness. All reviewed papers were concerned with either interpretability or explainability of AI models applied to weather forecasts [3]. It is especially important when the predictions generated by a particular model turn out to be inaccurate for some new extremes because then meteorologists must have an explanation for that.

Probabilistic calibration. The majority of reviewed papers evaluate only deterministic forecasts. Schulz and Lerch [28] demonstrate the value of calibrated probabilistic output for severe weather warnings, but this kind of evaluation remains the exception rather than the rule.

Developing regions and data equity. The large majority of reviewed papers use ERA5 or similar global reanalysis products, meaning they are fundamentally training on and evaluating in regions with dense observational networks. Dedicated work on Sub-Saharan Africa, small island states, and other data-sparse developing regions remains very limited [32].

8.4 Maturity Assessment by Application Area

Table 7 summarises our assessment of how mature different application areas are based on the evidence in the 101 reviewed papers.

Table: Maturity assessment of AI weather and climate forecasting application areas based on the 101 reviewed papers.

Application Area	Maturity Level	Evidence Papers	Supporting Evidence
Seasonal / S2S Forecasting	Nascent	5	Machine learning for seasons-ahead prediction is still at an early stage
Extreme Event Prediction	Active Research	6	Models show limited skill on rare, out-of-distribution events
Probabilistic AI Ensembles	Active Research	5	Calibrated ensemble generation remains an open challenge
Climate Downscaling	Competitive	3	Super-resolution CNN methods are well-established but not yet standard
Convection Parameterisation	Competitive	7	ML emulation works within training distribution; extrapolation is limited
Precipitation Nowcasting	Production-Ready	5	ConvLSTM, TrajGRU, and generative models outperform radar extrapolation
Post-processing & Calibration	Production-Ready	4	Neural network post-processing outperforms EMOS in multiple benchmarks
Global Medium-range Deterministic	Production-Ready	7	GraphCast, Pangu-Weather, FengWu exceed IFS HRES on standard metrics

9. Research Gaps and Future Directions (RQ5)

Based on systematic analysis of the 101 reviewed papers, we identify six specific research gaps that should guide the field over the next five years.

1. Dedicated extreme event benchmarks. WeatherBench 2 [27] provides excellent standardisation for average-condition forecasting but acknowledges that its evaluation period is too short to draw strong conclusions about rare extremes.An event-oriented benchmark, designed specifically for high-impact events (floods, heatwaves, cyclone intensifications), is badly needed. Such benchmark should define predetermined evaluation periods containing known extreme events and require probabilistic outputs.

2. Physics-aware model design. Literature overview indicates that incorporating physical constraints into neural networks leads to better performance and physical consistency. Examples include NeuralGCM [15] and the hybrid approach of nowcasting proposed by Das et al. [22]. Physics-aware design is a must for future studies aimed at modeling of weather/climate phenomena.

3. Interpretability methods specific to atmospheric sciences. Reviewing literature we could not find any mention of interpretability methods for machine learning models explaining the results to operational meteorologists [3]. New approaches and techniques are required to enable meteorologists to understand why a certain result was obtained, identifying important atmospheric processes and features involved in the forecast.

4. Regional coverage and access equity. It should be highlighted that almost all the models in review have been either trained or evaluated with ERA5 and other European/American-dominant datasets. Research efforts to predict South Asian monsoons [32] and African weather phenomena as well as to assist small island nations are severely underrepresented.

5. Probabilistic assessment using standardization. These statistics, namely CRPS, Brier scores, and reliability diagrams, must be made obligatory for reporting rather than being optional. This was clearly shown by Schulz and Lerch [28] when dealing with wind gusts post-processing.

6. Data fusion from multiple sources and modalities. Most foundation models train on gridded reanalysis datasets primarily. Combining multiple observation data streams – including ground station reports, lightning detection networks, radar reflectivities, satellite imagery, and crowdsourced observations – into one AI system is difficult but important from a scientific perspective. Mostajabi et al. [31] demonstrated that combinations of surface variables could be used for nowcasting locally, indicating that more intelligent fusion would provide significant gains in data-scarce areas.

10. Conclusion

A total of 101 articles related to AI applications in weather/climate prediction have been reviewed in this systematic review based on the PRISMA 2020 methodology among six leading databases over the years 2019 to 2024. This review employed clear guidelines as well as five research questions.

In regards to RQ1, there are currently three generations of architectures available within the scope. 2019-2021 saw the rise of recurrent network architectures like LSTM, GRU, and ConvLSTM. In contrast, 2022 marked the emergence of transformer models and graph neural networks. Foundation models, especially those trained on ERA5 or CMIP6 data sets, are seen as the future from 2023 onwards.

In the aspect of RQ2, we have found that AI weather forecasting models now consistently beat ECMWF IFS in medium-range deterministic benchmarking, with a massive difference in speed, from several hours of computation to seconds per forecast. Nonetheless, NWP models still possess unique benefits such as forecasting rare events and maintaining physical consistency under out-of-distribution scenarios as well as ensemble generation.

Regarding RQ4, geopotential height and temperature are the top predictors used for the task. The modern approaches employ hundreds of input predictors at once, whereas previous research usually limited themselves to selecting just a few of those.

Regarding RQ5, the most important open questions to answer would be: extreme events forecasting; physical consistency during distribution shifts; model interpretability; probabilistic calibration; equitable global coverage of developed nations and other parts of the world.

Given how quickly AI weather forecasts advance in terms of performance metrics, it is safe to say the future of such techniques looks bright. If the open questions mentioned above – physical consistency, extreme events, and interpretability in particular – are resolved within the next five years, AI weather predictions might actually end up being the backbone of forecasting infrastructure on Earth.

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